

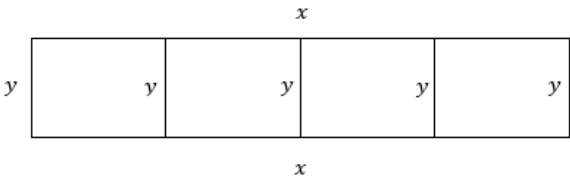
Algebra 1 Name Answer Key
 Unit 6 Date _____
 Lesson 8 Homework Period _____

Felix wants to build a rectangular fence in his yard that will be divided into *four* equal pens for chickens. Each pen needs to be the same size. He has 30 feet of fencing to build the fence. This scenario can be modeled as a *two*-variable inequality.

1. Describe what the two variables would represent in the context of Felix’s fence.

x will represent the length of the rectangle & y is the width of each section.

2. The diagram below shows a possible model of Felix’s fence with sides labeled as *x* and *y*.



3. Recall Felix had 30 feet of fencing to use. Write a *two*-variable inequality that Felix could use to find possible dimensions for his fence.

2x + 5y ≤ 30

4. Explain how to determine which inequality symbol to use.

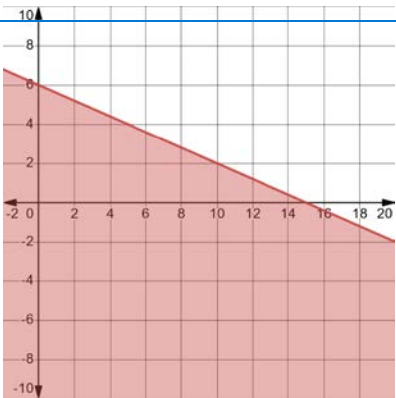
He only has 30 ft. so he can't use more than that.

5. Graph the inequality.

2x + 5y ≤ 30
−2x − 2x

5y ≤ −2x + 30
5 5 5

y ≤ −2/5x + 6



Commented [FAC1]: Link to graph: <https://www.desmos.com/calculator/avbeotbf9d>

Drew was applying for a personal loan to pay for home improvement projects. His bank requires a minimum of \$1,000 between his checking and savings accounts to be considered for a loan.

6. Define two variables for this situation.

Let x = checking account balance

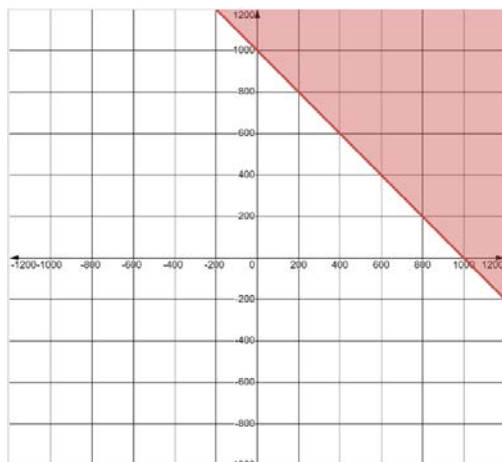
Let y = savings account balance

7. Write an inequality in terms of x and y that could be used to determine whether Drew meets the minimum requirement for the loan.

$$x + y \geq 1,000$$

8. Graph the inequality.

$$y \geq -x + 1000$$



9. Complete the statement.

To be a viable solution in this context, the x and y values must both be rounded to 2 decimal places.

10. Drew overdrew his checking account, which now has a balance of $-\$350.00$. He has $\$1,452.65$ in his savings account. Does he meet the minimum requirement?

$$\begin{array}{r} 1452.65 \\ -350.00 \\ \hline 1102.65 \end{array}$$

Yes, he still has at least \$1,000.